

CALDERON-ZYGMUND-STAMPACCHIA THEORY IN R^N , $N \geq 3$, $N=2$, FOR NONLINEAR DIRICHLET PROBLEMS (HAVING DISCONTINUOUS COEFFICIENTS)

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New Insights: Calculus of Variations, Optimization and Long-range Applications
[dedicated to Nicola Fusco]
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1. STARTING POINT: LINEAR CASE

In this talk Ω is a bounded, open subset of $\mathbb{R}^N$, with (now) $N \geq 3$, $f(x)$ belongs to $L^m(\Omega)$, with $m \geq 1$, $M(x)$ is a measurable, elliptic, bounded matrix.

Furthermore $m^* = \frac{mN}{N-m}$ (if $1 < m < N$) and $m^{**} = \frac{mN}{N-2m}$ (if $1 < m < N/2$). Let us consider the following boundary value problem

$$(1.1) \quad \begin{cases} -\operatorname{div}(M(x) \nabla u) = f(x) \in L^m(\Omega), & \text{in } \Omega; \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

1.1. CALDERON-ZYGMUND THEORY. If the coefficients of the matrix $M(x)$ are smooth, a possible presentation of the theory is:

$$(1.2) \quad f \in L^m(\Omega), \ 1 < m < N/2, \text{ implies } u \in W_0^{1,m^*}(\Omega), \text{ i.e. } Du \in L^{m^*}.$$

Then a consequence of the *Sobolev embedding* (here important difference between $N=2$ and $N \geq 3$) is

$$(1.3) \quad \text{Corollary : } f \in L^m(\Omega), \ 1 < m < N/2, \text{ implies } u \in L^{m^{**}}(\Omega).$$

1.2. CALDERON-ZYGMUND-STAMPACCHIA THEORY FOR LINEAR OPERATORS WITH DISCONTINUOUS COEFFICIENTS.

1.2.1. $\boxed{m > \frac{2N}{N+2}}$. If $f \in L^m(\Omega)$, $m > \frac{2N}{N+2}$, and the coefficients of $M(x)$ are discontinuous, the statement of (1.2) can fail (see [2], [12]).

Nevertheless, G. Stampacchia (see [16], with linear methods) proved some summability theorem of the solution $\mathbf{u}$.

1.2.2. $\boxed{1 \leq m < \frac{2N}{N+2}}$. On the other hand, if $f \in L^m(\Omega)$, $1 \leq m < \frac{2N}{N+2}$, we are outside of the finite energy solutions framework and the existence of a distributional solution u such that

$$(1.4) \quad \int_{\Omega} M(x) \nabla u \nabla \varphi = \int_{\Omega} f(x) \varphi(x), \quad \forall \varphi \in \mathcal{D}(\Omega),$$

is proved in joint papers with T. Gallouet ([6] and [7]), where it is proved that (infinite energy solutions)

$$(1.5) \quad \text{if } \begin{cases} 1 < m < 2N/(N+2), \\ m = 1, \end{cases} \quad \begin{cases} \text{then } u \in W_0^{1,m^*}(\Omega), \text{ as in (1.3);} \\ \text{then } u \in W_0^{1,q}(\Omega), \text{ } q < \frac{N}{N-1}. \end{cases}$$

2. CALDERON-ZYGMUND-STAMPACCHIA THEORY FOR NONLINEAR OPERATORS WITH NONREGULAR DATA

In the case of nonlinear differential operators of the type $A(v) = -\operatorname{div}(a(x, \nabla v))$ (Leray-Lions operator), under the classical assumptions on $a(x, \xi)$, the Dirichlet problem for finite energy solution is

$$(2.1) \quad u \in W_0^{1,p}(\Omega) : \int_{\Omega} a(x, \nabla u) \nabla v = \int_{\Omega} f(x) v(x), \quad \forall v \in W_0^{1,p}(\Omega).$$

If $f \in L^m(\Omega)$, $1 \leq m < (p^*)'$, the Calderon-Zygmund theory for linear operators with nonregular data is studied in some joint papers with T. Gallouet ([6], [7], [8]), where the existence of distributional solution is proved:

- if $\frac{N}{N(p-1)+1} < m < \frac{Np}{pN+p-N} = (p^*)'$, then there exists a distributional solution u of

$$(2.2) \quad u \in W_0^{1,q}(\Omega) : \int_{\Omega} a(x, \nabla u) \nabla \varphi = \int_{\Omega} f(x) \varphi(x), \quad \forall \varphi \in \mathcal{D}(\Omega),$$

where

$$(2.3) \quad q = (p-1)m^*;$$

- if $m = \frac{N}{N(p-1)+1}$, then there exists a distributional solution u of (2.2), and $u \in \mathbf{W}_0^{1,1}(\Omega)$; a radial example shows that $u \notin W_0^{1,q}(\Omega)$, for every $q > 1$;

2.1. Work in progress. “Minimization” (and Euler-Lagrange equation) problems of the type

$$\min \frac{1}{p} \int_{\Omega} |Dv|^p - \int_{\Omega} \rho(x) v_+^{\sigma}, \quad \sigma < p, \quad 0 \leq \rho(x) \in L^s(\Omega).$$

Dedicated to Nicola Fusco: for some s, p (*small*) there exists a solution $\mathbf{u}$ in $\mathbf{W}_0^{1,1}(\Omega)$, in the spirit of [8]. section SECTION N=2

- i:** All the previous results use the Sobolev inequality; in the N=2 framework the use of Trudinger inequality is less “smooth”.
- ii:** All the previous results depend on the summability of $f \in L^m(\Omega)$, with m from 1 to $m > N/2$. In the N=2 framework, the summability results depend on the summability of f in a microscopic scale: from $f \in L^1(\Omega)$ to $f \log(1+f) \in L^1(\Omega)$ (which implies $u \in L^\infty(\Omega)$, see [1]).
- iii:** Recent papers (nonlinear differential operators) in the spirit of (1.5) are [3] ($f \in L^1(\Omega)$: with a Brezis-Merle type estimate) and [4], where is proved that the assumption

$$\int_{\Omega} |f| [\log(1 + |f|)]^{\theta} \leq C_{\theta}, \quad 0 \leq \theta < 1,$$

implies

$$(2.4) \quad \int_{\Omega} e^{C^*|u|^{\frac{1}{1-\theta}}} \leq C_*, \quad (C^*(f, \theta), C_*(f, \theta)).$$

REMARK 2.1. *The case $\theta = 1$ (bounded solutions), studied in the past century in [1], is studied by a duality method in [4].*

REMARK 2.2. $-\Delta_N$ in $\mathbb{R}^N$: similar.

sectionSECTION N=2

2.2. Work in progress. Minimization (and Euler-Lagrange equation) problems of the type

$$\min \frac{1}{2} \int_{\Omega} M(x) Dv Dv - \frac{1}{1+\sigma} \int_{\Omega} \rho(x) v_+^{1+\sigma}, \quad \sigma < 1,$$

with $0 \leq \rho(x)$ and $\int_{\Omega} \rho(x) [\log(1 + \rho(x))]^{\theta} < \infty$, $1 < \theta \leq 1$

Dedicated to Nicola Fusco and Enzo Ferone, co-authors of the joint paper [BaFFO]=[5]: existence of solutions u (depending on θ).

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